

MICROWAVE AND PIR SENSORS: A Small Investment For Big Savings

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Microwave sensors offer great advantages in security and energy savings with their high sensitivity

Microwave sensors are cost-effective solutions to save energy expenses and secure premises of businesses. Their most popular use is in lighting, where these can save up to 40 per cent of electricity expenditure for companies.

Devices containing microwave sensors generate microwave rays, which cover a certain area across the space. When a person walks into the microwave zone, his body causes the waves to reflect, which is detected by the sensor for further action. Occupancy sensors, in particular, are quite popular. These use infrared or microwave technology depending on the application.

Let's explore how microwave sensors benefit businesses.

Cuts energy bills by half

Electric appliance control is a popular use case of microwave sensors. Microwave sensors can cut a substantial amount of energy wastage by automatically turning the lights off when an area is unoccupied.

Lal Changlani, director, Sensinova Lighting, comments, "Microwave sensors can save 30-80 per cent of electricity expenses depending on the frequency of motion in the area and the mode of energy usage. Areas with more intermittent occupancy will generate more savings. For instance, corridors and staircases have intermittent traffic throughout the day. The time during which these areas remain unoccupied will generate complete savings. Additionally, many places have lights running in staircases and corridors even at night between 10pm and 5am. These expenses can be completely cut down with automated control. Nowadays, microwave sensors enable auto-dimming of lights, further optimising the energy charts for businesses."

Rajath Rao, managing partner, Ecomica Technologies, shares an example of microwave sensors usage: Kuwait-based Baiti Synergy was experiencing high electricity costs for its office space. The main reason behind excessive energy consumption was office cubicles being lighted even when these were unoccupied—both during and after office hours. Baiti installed microwave sensor solutions from Ecomica in office cubicles, cabins and other areas of its workspace. The sensors themselves consumed 1.1 watts, running 24 hours a day. Energy audit after a few electricity billing cycles revealed a steady cost reduction by 30 per cent and very quick payback on the investment.

Provides tight security

Security applications require very sensitive devices to pick up even the slightest of movements in closed areas. The consistent emission of microwave rays makes it impossible for intruders to pass undetected. An active coverage of up to 120 metres sets off the alarm the moment thieves, burglars and intruders enter the premises.

Rajath explains, "Microwave sensors have heightened sensitivity due to the presence of fewer dead zones in their detection field. Hence these are used for detection of very little movements. They find a lot of use in security applications such as intrusion alarm systems for closed-environment spaces of offices."

A spokesperson from Pyrotech Electronics shares, "Microwave sensors detect any motion even beyond walls or behind doors. Warehouses with restricted-access areas, banks or rooms with lockers use microwave sensors for the same reason. Any motion across the restricted area, even if it is behind doors, gets detected and the alarm sets off."

Industrial applications. Conventional sensor solutions to measure the performance of industrial devices are quite expensive. Additionally, these require regular maintenance and adjustments, resulting in some downtime. Microwave sensors provide benefits like high interference immunity, one-time servicing and tuning for lifetime operation, high precision and high reli-

Fig. 1: A microwave sensor

